UNIT – 1	9 Hours
Graph Theory in Electrical Circuits	
Graph Theory in Electrical Circuits: Ohm's law, Kirchhoff Laws. Graph Representation: Nodes, Branches, Linear graph of a network. Tie set matrix and Cut-set Matrix , Analysis of resistive network using cut -set and tie set.	

Introduction

The interconnection of various electric elements in a prescribed manner comprises as an electric circuit in order to perform a desired function. The electric elements include controlled and uncontrolled source of energy, resistors, capacitors, inductors, etc. Analysis of electric circuits refers to computations required to determine the unknown quantities such as voltage, current and power associated with one or more elements in the circuit. To contribute to the solution of engineering problems one must acquire the basic knowledge of electric circuit analysis and laws. In electrical engineering, solution of network analysis problems involves finding the current through and voltage across different circuit elements. The graph theory deals with the properties of networks which do not change with the change in the shape of the networks. The graph theory concept eases the solution method for solving networks with a large number of nodes and branches.

1.1 Ohms Law: Ohm's law states that the voltage across a conductor is directly proportional to the current flowing through it, provided all physical conditions and temperatures remain constant.

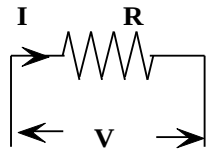


Fig 1.1 Electric Circuit

$$I = \frac{V}{R}$$

1.2 KIRCHOFF LAWS

Gustav Kirchhoff, a German physicist, observed that these were instances of two general conditions fundamental to the analysis of any electrical network. These conditions may be stated as follows:

1.2.1 KIRCHOFF'S CURRENT LAW

Kirchhoff's Current Law (KCL) states that the total current entering a node (junction) in an

electrical circuit is equal to the total current leaving that node. This is also known as Kirchhoff's first law or the junction rule. Essentially, it reflects the principle of conservation of charge, meaning that charge is not lost or gained within a circuit.

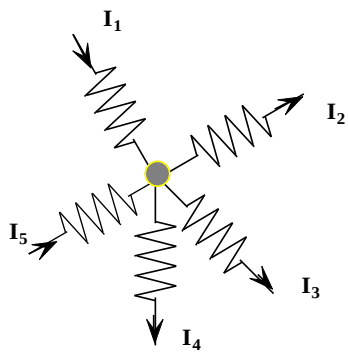


Fig 1.2 Electric Circuit

Convention: Current entering a node -ve, leaving the node +ve

-I1+I2+I3+I4-I5=0

i.e. I2+I3+I4=I1+I5

Exercise 1: Find the magnitude and direction of the unknown currents in Fig 1.3 Given i_1=10A, i_2=6A and i_5=5A

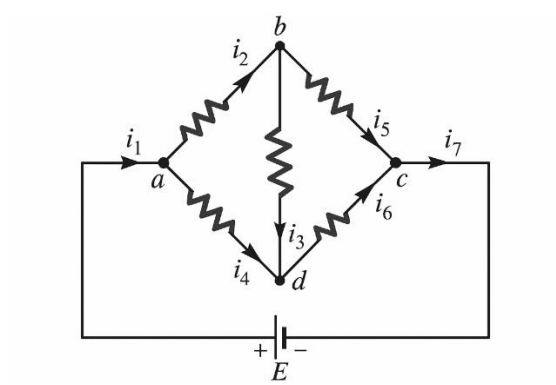


Fig.1.3

By Applying KCL $i_1 = i_7$ i.e. $i_7 = 10A$

At node "a" $i_1 = i_2 + i_4$

$10 = 6 + i_4$, hence $i_4 = 4A$

At node "b", Applying KCL

$i_2 = i_3 + i_5$

or $i_3 = i_2 - i_5 = 6 – 5 = 2A$

At node "c" $i_7 = i_5 + i_6$

$i_6 = i_7 - i_5 = 10 – 4 = 6A$

Hence $i_1 = i_7 = 10A$

$i_2 = 6A$

$i_3 = 2A$

$i_4 = 4A$

$i_5 = 4A$

$i_6 = 6A$

Note: The directions of currents can be arbitrarily assigned by applying KCL and can be changed if the obtained values are negative.

1.2.2 KIRCHOFF'S VOLTAGE LAW

Kirchhoff's Voltage Law (KVL) states that the sum of all voltage drops around any closed loop in a circuit is equal to zero. This law is a consequence of energy conservation in electrical circuits. In simpler terms, the total voltage source (like a battery) in a loop must equal the total voltage drops across the components in that loop. Consider the closed electric circuit shown in Fig 1.4

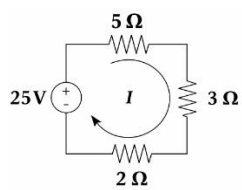


Fig 1.4 Electric Circuit

Convention: Voltage drops -ve. Source – to + is + ve , + to – is -ve

Hence applying to KVL to the circuit shown in Fig.1.4

$$-5I-3I-2I+25=0$$

$$10I=25$$

$$\mathbf{I=2.5\ A}$$

1.2.3 Power

Power is an indication of how much work (the conversion of energy from one form to another) can be done in a specified amount of time, that is, a rate of doing work. The electrical unit of measurement for power is the watt (W), defined by 1watt (W)=1joule/second (J/s).

$$P = VI = V\frac{V}{R} = \frac{V^2}{R}W$$

$$P = VI = (IR)I = I^2R\ W$$

Exercise 2: Apply KVL for the network shown in Fig. 1.5, find the power absorbed by 18 Ω resistor and power supplied by 10V voltage source.

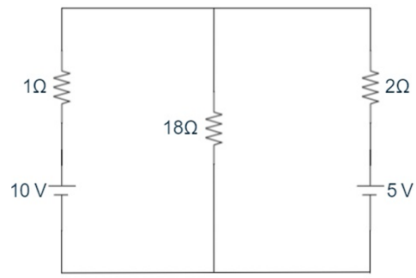
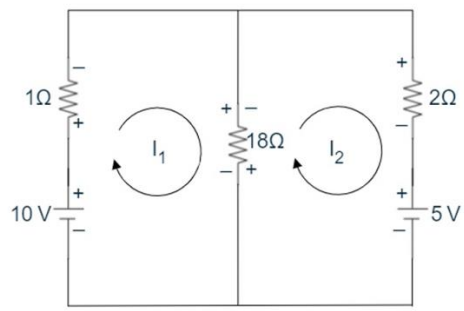


Fig 1.5. Electric Circuit

Solution:



Loop 1: $10 - 1(I_1) - 1(I_1) - 18(I_1 - I_2) = 0$

$$19I_1 + 18I_2 = -10$$

Loop 2: $-18I_2 + 18(I_1) - 2(I_2) - 5 = 0$

$$18I_1 - 20(I_2) = 5$$

By solving above equations, we get

$$I_1 = 1.9642A \; and \; I_2 = 1.5178A$$

The current through the 10V source and 1 Ω resistor is 1.9642A and the current through the 5V source and 2 Ω resistor is 1.5178A. The current through the 18 Ω resistor is determined by the following equation from the original network: $I_{18\Omega} = \mathrm{I}_1 - \mathrm{I}_2 = 1.9642\mathrm{A} - 1.5178\mathrm{A} = 0.446A$

Power absorbed by 18 Ω = I^2X18 = 3.58691W

Power supplied by 10 V battery = 1.9642X10 = 19.642W

1.2.4 Mesh analysis or Loop analysis of electric circuit with Independent Sources

The mesh analysis, the name being derived from the similarities in appearance between the closed loops of the network and a physical "fence" or mesh. The current through a mesh is known as mesh current. In this method, a distinct current is assumed in the loop and the

polarities of drops in each element in the loop are determined by the assumed direction of loop current for the loop. KVL is then applied around each closed loop and by solving this loop equation, the branch currents are determined.

The following steps are used in Mesh analysis

Identify the meshes in an Electric Circuit. Consider the circuit shown in Fig 1.6

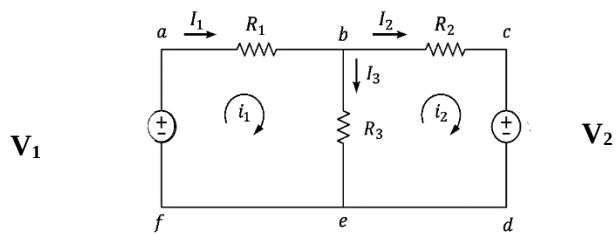


Fig 1.6

There are two meshes:

Mesh 1: abefa

Mesh 2: bcdeb

Assign the loop currents. Current i_1 flows in loop 1 and current i_2 flows in loop 2.

Apply KVL to each loop

Consider loop1:

$-I_1R_1-(I_1-I_2)\ R_3+V_1=0$

$-(R_1+R_3)\ I_1+R_3I_2=-V_1$

$(R_1+R_3)\ I_1-R_3I_2=V_1$ (3)

Consider loop2:

$-I_2R_2-V_2-(I_2-I_1)\ R_3=0$

$R_3I_1-(R_1+R_3)\ I_2=V_2$ (4)

Solve simultaneous equations (3 and (4) to find I_1 and I_2

Exercise 3: For the circuit shown in Fig. 1.7, find the branch currents, using mesh analysis.

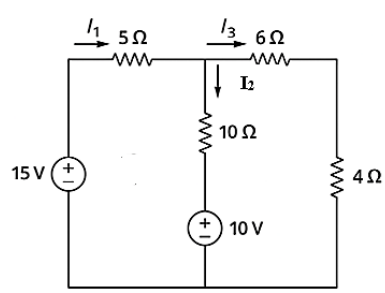


Fig. 1.7

Write KVL to Mesh 1,

$$15 - 15I_1 - 10\,(I_1 - I_2) = 0$$

$$-15I_1 + 10\,I_2 = -15 \qquad (1)$$

Write KVL to Mesh 2,

$$-10 - 10\,((I_2 - I_1) - 6I_2 - 4I_2 = 0$$

$$10I_1 - 20I_2 = 10 \qquad (2)$$

Solving (1) and (2),

$$I_1 = 1A$$

$$I_2 = 0\ A$$

Hence, $I_3 = I_1 - I_2$

$$I_3 = 1\ A$$

1.2.5 Nodal analysis of electric circuit with Independent Sources

Nodal analysis is a method that provides a general procedure for analysing the circuits using node voltages as the circuit variables. It is also called Node-Voltage Method.

In the nodal method, the number of independent node-pair equations needed is one less than the number of junctions in the network. If "n" denotes the number of independent node equations of "j" the number of junctions, then n = j – 1

The following steps are used in Node analysis

Identify the nodes in an Electric Circuit. Consider the circuit shown in Fig.1.8

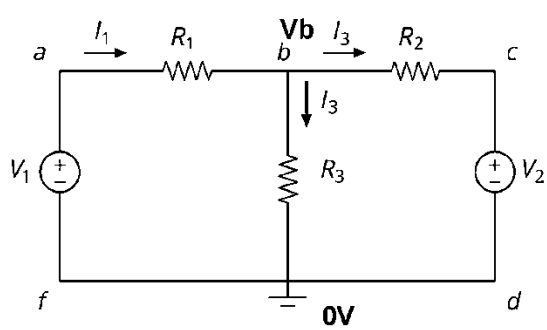


Fig.1.8

There are two nodes:

Node 1: b

Node 2: e

Assign the node voltages at different nodes. Voltage V_b at node b and 0V at node e (datum node)

Apply KCL at node b

$$\frac{(V_b - V_1 - 0)}{R_1} + \frac{(V_b - 0)}{R_3} + \frac{(V_b - V_2 - 0)}{R_2} = 0$$

Solve for V_b

Compared with the mesh method (where KVL equations are frequently used to solve the network), the nodal method is advantageous when the network has many parallel circuits; otherwise, both the nodal and mesh methods offer almost equal advantages for the mesh method the number of independent mesh equations needed is m = b – (j – 1), where b is the number of branches.

Note: If m < n the mesh method is preferred and m > n, nodal method is preferred.

Exercise 4: Using Nodal method, find the current through r_2 shown in Fig.1.9

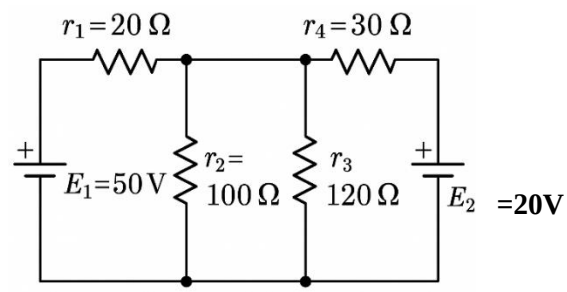


Fig.1.9

Solution: The circuit can be redrawn and written as shown in Fig.1.10 with naming the nodes and branch currents.

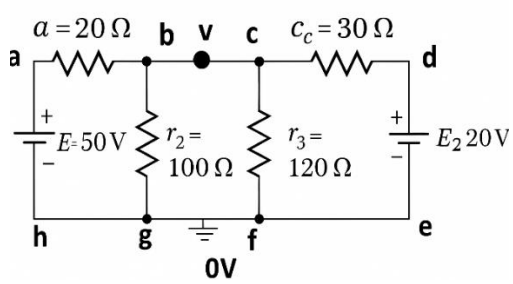


Fig.1.10

From the Figure, $V_c = V_b = v$(Node b and c are at the same potential)

Applying KCL at node b, $\frac{(v-50)}{20} + \frac{(v-0)}{100} + \frac{(v-0)}{120} + \frac{v-20}{30} = 0$

Solving the equation for v , v = 31.18 V

$$i_2 = \frac{v}{r_2} = \frac{31.18}{100}\ 0.3118\ A$$

The current through r_2 = 311.8 mA

1.3 Graph representation

The graph of a network is drawn by first marking the nodes and then joining these nodes by lines which correspond to the network elements of each branch. All the voltage and current sources are replaced by their internal impedances. The voltage sources are replaced by short circuits as their internal impedances are zero whereas current sources are replaced by open circuits as their internal impedances are infinite.

1.3.1 Terminology used in the network graph

Node

It is defined as a point at which two or more elements have a common connection.

Example: a, b, c, d, e, f and g

Branch

It is a line connecting a pair of nodes, the line representing a single element or series connected elements. Example: v_1, R_1, R_2, R_3, v_2, R_4, R_5, R_6, R_7 and I

Loop

It is the closed contour selected in a graph. Example: abedca, abegfca, cdebgfc

Mesh

It is a loop which does not contain any other loops within it.

Example: abedca, cdegfc, gebg (through R_7) and gebg (through I)

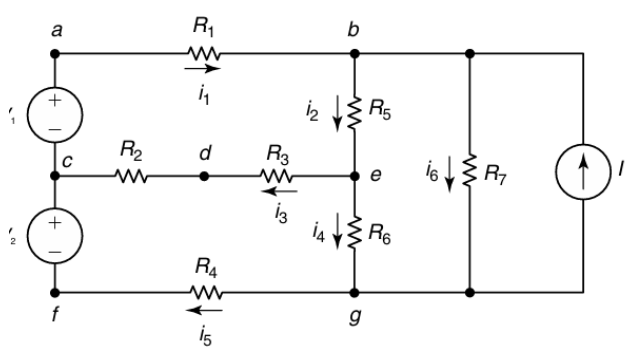


Fig 1.11 Circuit illustrating terminologies

1.3.2 Linear graph of a network

A linear graph is a collection of nodes and branches. The nodes are joined together by branches. Figure 1.11 shows a network and its associated graphs. Each branch of a graph may be given an

orientation or a direction with the help of an arrow head which represents the assigned reference direction for current. Such a graph is then referred to as a directed or oriented graph. Branches whose ends fall on a node are said to be incident at that node. Degree of a node is defined as the number of branches incident to it. Branches 2, 3 and 4 are incident at Node 2 in Fig. 1.12 (c). Hence, the degree of Node 2 is 3.

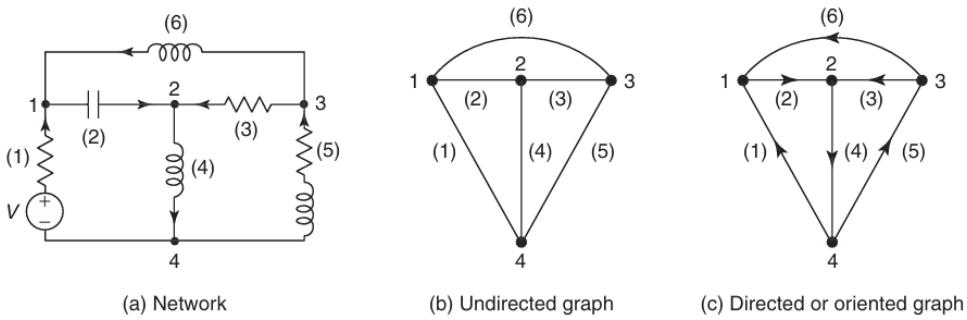


Fig 1.12. Network and its graphs

Tree

It is an interconnected open set of branches which include all the nodes of the given graph. In a tree of the graph there cannot be any closed loop.

Tree branch or Twig

It is an any branch of a tree. The number of twigs on a tree is always one less than the number of nodes. Let N= Number of nodes; then number of twigs = N -1

Tree link or chord

It is that branch of the graph that does not belong to the particular tree.

The number of links (L) = B – N + 1

Where,B = Number of branches , N = Number of nodes

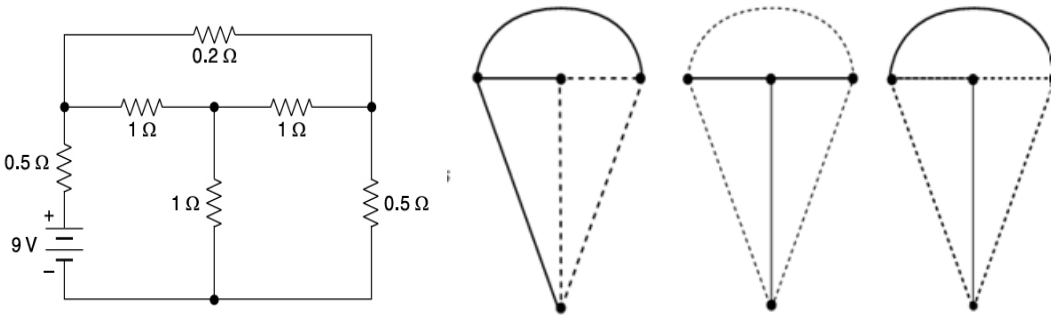


Fig 1.13. Circuit and possible trees of the circuit

1.3.3 Tie-Set:

For a given tree of a graph, addition of each link between any two nodes forms a loop called the

fundamental loop. In a loop there exists a closed path and a circulating current, which is called the link current. The current in any branch of a graph can be found by using link currents.

The fundamental loop formed by one link has a unique path in the tree joining the two nodes of the link. This loop is also called **f-loop** or a **tie-set**.

A **tie-set** is a set of branches contained in a loop such that each loop contains one link or chord and the remainder are tree branches.

Consider the graph and the tree as shown in Figure1.14(a) and Figure1.14(b). It has four nodes and six branches. One of its trees is arbitrarily chosen and is shown in Figure1.14(b).

The twigs of this tree are branches 4, 5, and 6.

The links corresponding to this tree are branches 1, 2, and 3.

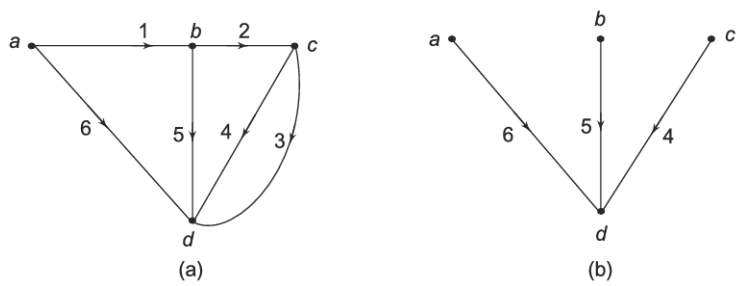


Figure 1.14: (a):Graph, (b):Tree

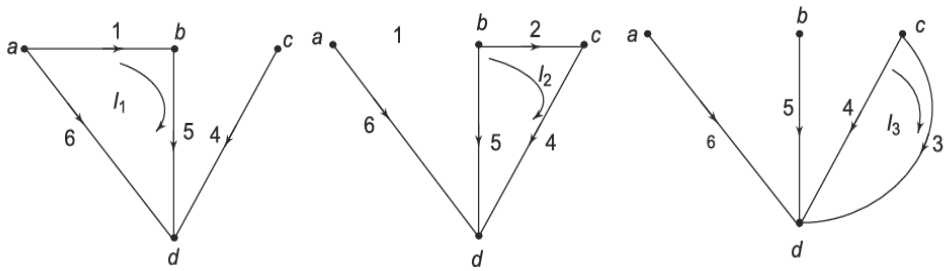
Number of nodes: n=4

Number of branches: b= 6

Number of tree branches or twigs = n-1= 3

Number of link branches I=b-(n-1)=3

This selected tree will result in three fundamental loops as we connect each link, in turn to the tree. The fundamental loops are shown in Figure 1.15.



(a) (b) (c)

Figure 1.15:(a) Loop 1, (b) Loop 2, (c) Loop 3

1.3.4 Tie-Set Matrix (B_a)

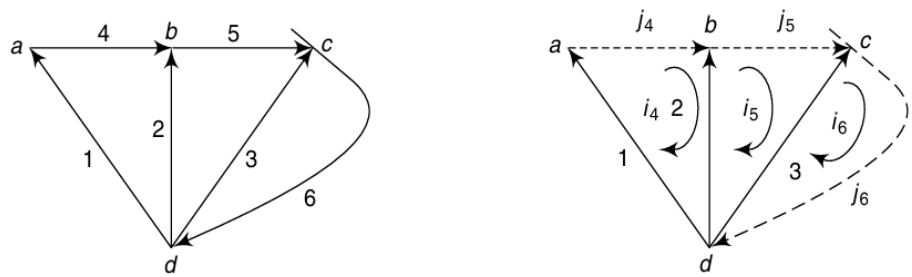
For a given graph having n nodes and b branches, tie-set matrix is a rectangular matrix with b columns and as many rows as there are loops.

Its elements have the following values

$B_{ij}=1$ if branch j is in loop i and their orientations coincide, that is, the loop current and the branch current flow in the same direction,

$B_{ij} = -1$ if branch j is in loop i and their orientations do not coincide.

$B_{ij} = 0$ if branch j is not in loop i.



(a) **(b)**

Figure 1.16: (a) Graph, (b)Selected Tree

For the graph shown in Figure 1.16(a) and tree selected in Figure 1.16(b) the tie-set matrix is written as follows.

	Branch no. (i)					
Links (j)	1	2	3	4	5	6
4	1	–1	0	1	0	0
5	0	1	–1	0	1	0
6	0	0	1	0	0	1

The entries in the Tie-set schedule are given as +1 or 1 if the branch current is in the same direction as the link current or not. If the branch current does not depend on the link current, then entry is zero.

1.3.5 Cut-Set:

A cut-set is a minimum set of elements that when cut, or removed, separates the graph into two groups of nodes. A cut-set is a minimum set of branches of a connected graph, such that the removal of these branches from the graph reduces the rank of the graph by one.

Consider the graph shown in Figure 1.17(a). The rank of the graph is 3. The removal of branches 1 and 3 reduces the graph into two connected subgraphs as shown in Figure 1.17(b-c).

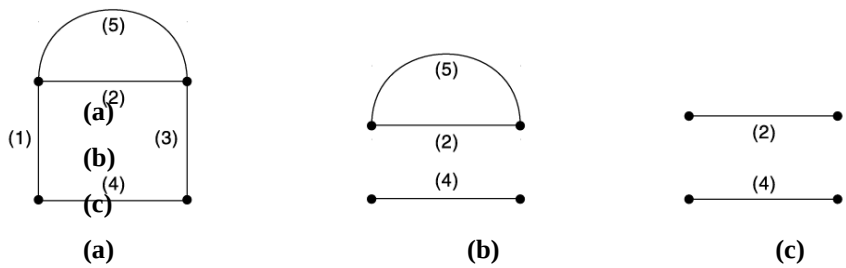


Figure 1.17: (a) Graph, (b) Subgraphs with removal of 1 and 3, and (c) Subgraphs with removal of 1, 3 and 5.

- The rank of the graph of Figure 1.17(a) = (4 -1) = 3
- The rank of the subgraph of Figure 1.17(b) = Addition of the ranks of the subgraphs= 1+1=2. So, branches [1, 3] may be a cut-set.
- Also, removal of the branches 1, 3 and 5 reduces the graph into two connected subgraphs as shown in Figure 1.17(c) and the rank becomes 2. So, [1, 3, 5] may also be a cut-set.

As cut-set is the minimum set of branches and [1, 3] is a subset of [1, 3, 5], so [1, 3] is the cut-set, [1, 3, 5] is not a cut-set.

Fundamental Cut-Set: A fundamental cut-set (FCS) is a cut-set that cuts or contains one and only one tree branch. Therefore, for a given tree, the number of fundamental cut-sets will be equal to the number of twigs.

Procedure for Finding the Fundamental Cut-sets:

- First, select a tree of the given graph
- Focus on a tree branch
- Check whether removing this tree branch from the tree disconnects the tree into two separate parts.
- All the links which go from one part of this disconnected tree to the other, together with the tree branch forms a fundamental cut-set.

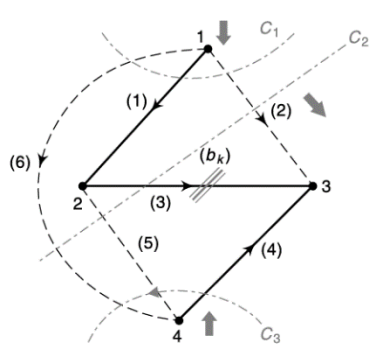


Figure 1.18: Graph illustrating fundamental cut-set

Following this procedure, the fundamental cut-sets for the graph of Figure 1 will be

- f-cut-set 1: [1, 2, 6];
- f-cut-set 2: [2, 3, 5, 6];
- f-cut-set 3: [4, 5, 6]

Properties of Cut-Set:

- A cut-set divides the set of nodes into two subsets.
- Each fundamental cut-set contains one tree-branch, the remaining elements being links.
- Each branch of the cut-set has one of its terminals incidents at a node in one subset and its other terminal at a node in the other subset.
- A cut-set is oriented by selecting an orientation from one of the two parts to the other. Generally, the direction of cut-set is chosen same as the direction of the tree branch.

1.3.6 Cut-Set Matrix (Q_C):

For a given graph, a cut-set matrix (Q_C) is defined as a rectangular matrix whose rows correspond to cut-sets and columns correspond to the branches of the graph. Its elements have the following values:

$$\begin{aligned} Q_{ij} &= 1, \text{ if branch } j \text{ is in the cut-set } i \text{ and the orientations coincide.} \\ &= -1, \text{ if branch } j \text{ is in the cut-set } i \text{ and the orientations do not coincide.} \\ &= 0, \text{ if branch } j \text{ is not in the cut-set } i. \end{aligned}$$

For the graph shown in Figure 1.18, fundamental cut-sets have been identified as follows.

	Branch no:					
f-cut-sets	1	2	3	4	5	6
1	1	1	0	0	0	1
2	0	1	1	0	1	1
3	0	0	0	1	–1	–1

1.3.7 Incidence Matrix:

The incidence matrix symbolically describes a network. It also facilitates the testing and identification of the independent variables. Incidence matrix is a matrix which represents a graph uniquely.

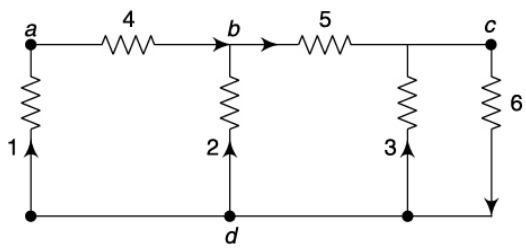
This matrix tells us which branches are incident at which nodes and what are the orientations relative to the nodes.

For a given graph with n nodes and b branches, the complete incidence matrix A_a matrix of order n×b, whose elements have the following values.

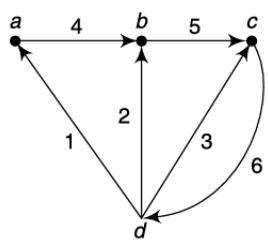
A_{ij} = 1, if branch j is associated with node i and oriented away from node i.
= –1, if branch j is associated with node i and oriented towards node i.
= 0, if branch j is not associated with node i.

The matrix obtained from $\mathbf{A_a}$ by eliminating one of the rows is called Reduced Incidence Matrix **A**. In other words, suppression of the datum node (reference node) from the incidence matrix results in reduced incidence matrix.

Example: Find the incident matrix for the network.



The graph of the matrix is



Complete Incidence matrix is

		Branches 1	2	3	4	5	6	
Nodes	a	−1	0	0	+1	0	0	Reduced Incidence Matrix A
	b	0	−1	0	−1	+1	0	
	c	0	0	−1	0	−1	+1	
Reference Node	d	+1	+1	+1	0	0	−1	

1.3.8 Analysis of resistive network using tie-set:

Consider the electrical network shown in Figure 1.19. The corresponding oriented graph of the network is shown in Figure1.19(a). From the oriented graph, a tree is selected as shown in the Figure1.19(b).

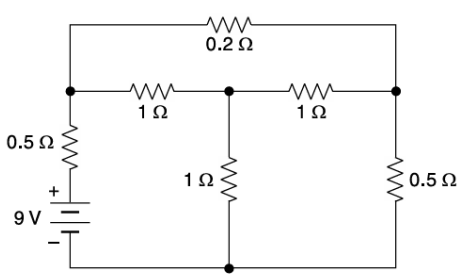


Figure 1.19

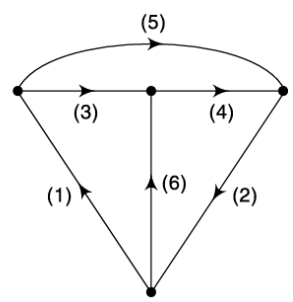


Figure 1.19(a)

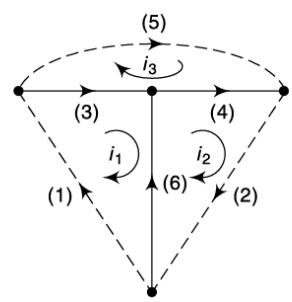


Figure 1.19(b)

The total number of branches in the network is $b = 6$

The number of nodes is $n = 4$

Hence, the number of links is given by $l = b - (n - 1) = 6 - 3 = 3$

Since each link produces one independent tie-set, the network has three tie-sets. Therefore, the tie-set matrix is of order 3×6

The tie-set matrix for the selected tree and assumed branch orientations is given by:

$$B_a = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & -1 & 1 & 0 \end{bmatrix}$$

1.3.9 Analysis of resistive network using cut -set:

Consider the electrical network shown in Figure 1.20. The graph and a suitable tree for the network are shown in the Figure 1.21.

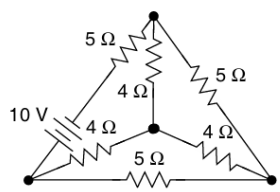


Figure 1.20

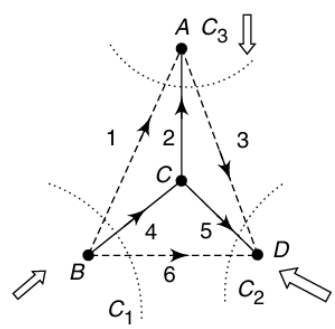


Figure 1.21

For the above network:

Total number of branches $b = 6$

Total number of nodes $n = 4$

Number of twigs (tree branches) $n - 1 = 3$

Number of links $l = b - (n - 1) = 6 - 3 = 3$

Each twig gives rise to one fundamental cut-set. Hence, the network has three fundamental cut-sets. Since the number of twigs is three and the total number of branches is six, the order of the cut-set matrix is:

$$3 \times 6$$

The fundamental cutsets are identified as,

f-cutset-1: [1, 4, 6]

f-cutset-2: [3, 5, 6]

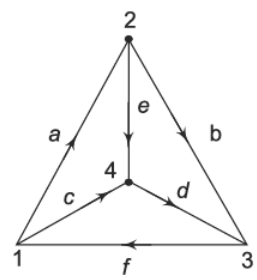
f-cutset-3: [1, 2, 3]

The fundamental cut set matrix is given as,

$Q =$

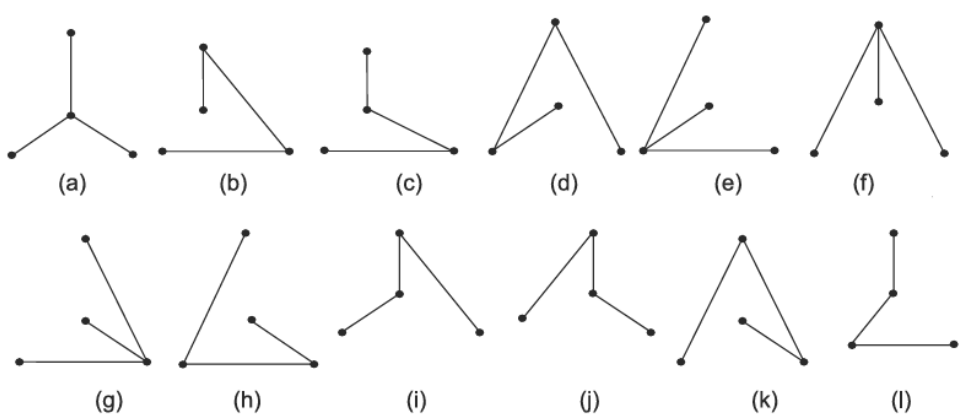
	1	2	3	4	5	6
C_1	−1	0	0	1	0	1
C_2	0	0	−1	0	−1	−1
C_3	−1	−1	1	0	0	0

Exercise 1. For the given graph shown below, draw the number of possible trees.

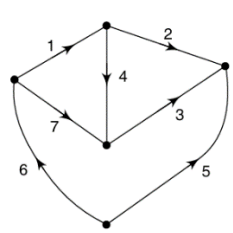


Solution:

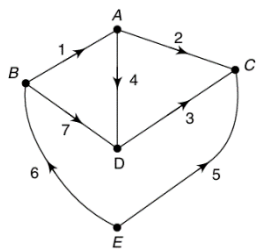
The number of possible trees are represented as follow:



Exercise 2. Write the complete incidence matrix for the graph shown in the figure.



Solution:



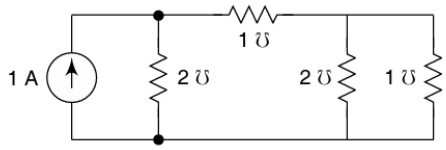
The graph is labeled as follow:

The complete incidence matrix is given as,

$A_a =$

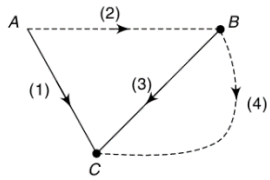
	1	2	3	4	5	6	7
A	−1	1	0	1	0	0	0
B	1	0	0	0	0	−1	1
C	0	−1	−1	0	−1	0	0
D	0	0	1	−1	0	0	−1
E	0	0	0	0	1	1	0

Exercise 3. For the network shown in the below figure, find the fundamental cutset matrix.



Solution:

The graph and one tree are shown for the network.



The fundamental cutsets are identified as:

f-cutset-1: [1, 2]

f-cutset-2: [2, 3, 4]

The fundamental cutset matrix is given as,

$Q_a =$

	1	2	3	4
C_1	1	1	0	0
C_2	0	−1	1	1

Exercise 4. The fundamental cut-set matrix is given as,

Twigs				*Links*		
1	2	3	4	5	6	7
1	0	0	0	–1	0	0
0	1	0	0	1	0	1
0	0	1	0	0	1	1
0	0	0	1	0	1	0

Draw the oriented graph of the network.

Solution:

The graph has seven branches and four fundamental cut-sets:

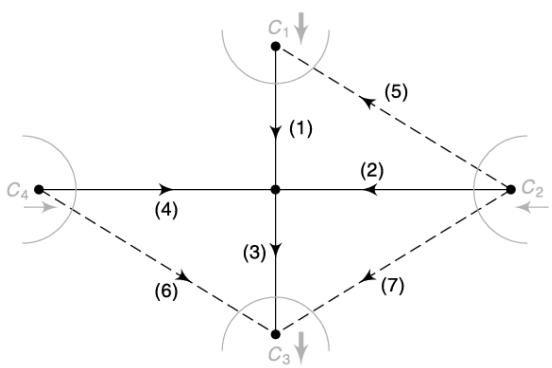
Cut-set-1: [1, 5]

Cut-set-2: [2, 5, 7]

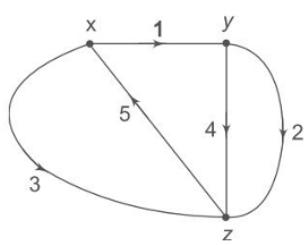
Cut-set-3: [3, 6, 7]

Cut-set-4: [4, 6]

So, the oriented graph is as shown in below figure.



Exercise 5 . For the graph shown below, find the incidence and tie-set matrix.



Solution:

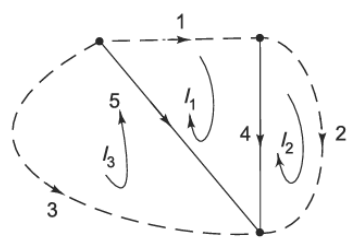
The nodes are x, y, and z and branches are 1, 2, 3, 4, and 5. The complete incidence matrix is given by

$$
\begin{array}{cc}
\text{nodes} & \text{branches} \rightarrow \\
\downarrow & \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\
A = \begin{array}{c} x \\ y \\ z \end{array} & \begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ -1 & 1 & 0 & 1 & 0 \\ 0 & -1 & -1 & -1 & 1 \end{bmatrix}
\end{array}
$$

Let us choose the node z as the reference or datum node for writing the reduced incidence matrix A_1.

$$
\begin{array}{cc}
\text{nodes} & \text{branches} \rightarrow \\
\downarrow & \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\
A_1 = \begin{array}{c} x \\ y \end{array} & \begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ -1 & 1 & 0 & 1 & 0 \end{bmatrix}
\end{array}
$$

For writing the tie-set matrix, consider the tree in the graph



Number of nodes,

$$n = 3$$

Number of branches,

$$b = 5$$

Number of tree branches (twigs),

$$\text{Twigs} = n - 1 = 3 - 1 = 2$$

Number of link branches (links),

$$l = b - (n - 1) = 5 - (3 - 1) = 3$$

The tie-set matrix or fundamental loop matrix is given by

$$
\begin{array}{cc}
\text{loop} & \text{branches} \rightarrow \\
\downarrow & \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\
B = \begin{array}{c} I_1 \\ I_2 \\ I_3 \end{array} & \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}
\end{array}
$$